THE GREEN USER
DESIGN FOR SUSTAINABLE BEHAVIOUR

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ABSTRACT

To reduce the environmental impact of the use phase of products, researchers have suggested applying design strategies for sustainable behaviour. The aim of this study was to evaluate four different design strategies on the basis of long-term acceptability and effectiveness in inducing sustainable behaviours. A literature review was carried out and a model for categorisation of strategies was created containing five categories: *Enlighten*, *Spur*, *Steer*, *Force* and *Match*. Four design strategies for sustainable behaviour, belonging to the first four categories, were implemented in prototypes to achieve moderate dosing of washing detergent. The prototypes were distributed to 16 households and a between subject study design was applied. The results indicate that three of the four strategies for many households were both effective and accepted. This suggests that product design can be a feasible way to induce and maintain sustainable behaviour.

Keywords: eco-design strategies, sustainable behaviour, washing detergent

INTRODUCTION

For some products, the use phase contributes the most to the total environmental impact and many ways to reduce the impact from this phase have been suggested, e.g. increased energy efficiency and lightweight constructions. Yet, mostly matters beyond the user's control have been investigated but accomplished efficiency gains do not always make a great difference due to unsustainable use patterns (Tang and Bhamra, 2009). To further reduce the impact of the use phase matters within the user's control should be addressed as well.

It has been suggested that designers can reduce the impact of use by applying design strategies for sustainable behaviour (Lilley, 2009), where the term 'sustainable behaviour' denotes using a product or service in a way that generates a smaller impact on the environment than conventional ways of using similar products or services. However, few of these design strategies for sustainable behaviour have been empirically evaluated. The aim of this study was therefore to examine the long-term effectiveness and acceptability of four different design strategies.

The behaviour on which to test the four design strategies was overdosing of washing detergent. Dosing of washing detergent by estimations, instead of with measuring devices, has been found to be a common consumer behaviour that may cause overdosing (Järvi and Pavoliita, 2007), which in turn causes unnecessary strain to the wastewater systems and may pollute watercourses. Overdosing of washing detergent was considered to have an intermediate resistance to change since it often is a habitual behaviour, yet has limited influence on other areas of everyday life. Moderate dosing of washing detergent was thus defined as the desired sustainable behaviour.

APPROACH

In a first phase literature was reviewed to identify promising design strategies. In the second phase the topic of dosing of washing detergent was explored to establish possible barriers towards moderate dosing. Four design strategies for sustainable behaviour were chosen to match the barriers. In a third phase, these strategies were applied in product prototypes. In the fourth phase the prototypes were distributed to households for evaluation.

PHASE 1: IDENTIFYING DESIGN STRATEGIES FOR SUSTAINABLE BEHAVIOUR

Researchers have suggested that product and service design can be used to influence behaviours to become more sustainable (e.g. Lockton, Harrison and Stanton, 2008; Lilley, 2009; Dwyer et al. 1993; Wever, van Kuijk and Boks, 2008; Jelsma and Knot, 2002). A notable collection of intervention strategies for behavioural change was for instance made by Bhamra, Lilley and Tang (2008) - containing *Eco-Information*, *Eco-Choice*, *Eco-Feedback*, *Eco-Spur*, *Eco-Steer*, *Eco-Technical Intervention* and *Clever Design*. Another set of design strategies for sustainable behaviour was suggested by Wever, van Kuijk and Boks (2008) and presented with a distinction between the strategies *Eco-Feedback*, *Scripting* and *Forced Functionality* that all requires behaviour adaption and the strategy *Functionality Matching* that does not.

The design strategies for sustainable behaviour mentioned and additional strategies found in the studied literature were compiled and a model for categorisation of the design strategies was developed. The complete compilation can be found in Lidman and Renström (2011).

A MODEL FOR CATEGORISATION OF DESIGN STRATEGIES

The model includes five strategy categories denominated *Enlighten*, *Spur*, *Steer*, *Force*, and *Match*. The first four categories – *Enlighten*, *Spur*, *Steer* and *Force* – require the user to change his or her behaviour. The two basic influencing techniques are to motivate the user, or to direct the user. These four categories are positioned with respect to the level of 'user control' versus 'designer control'. The fifth category, *Match*, differs from the other categories, as it requires none or very little adaptation from the user. In the category *Match*, both the user and the designer are in control since the user controls his or her behaviour, but the designer controls the outcome of the behaviour (Figure 1).

The purpose of the design strategies brought together in the category *Enlighten* is to motivate sustainable behaviours by influencing people's knowledge, values, attitudes and norms. This can be done through for instance information, feedback or means for reflection.

In the category *Spur*, all design strategies have in common that they encourage and tempt the user to perform the desired behaviour by means apart from the positive environmental consequences of the behaviour. The focus is either on other positive consequences of the behaviour or on the behaviour itself. This can be done through e.g. incentives or goal setting.

The design strategies within the category *Steer* guide the user by making a sustainable behaviour the evident choice. This could be done physically or cognitively by for instance constraints and affordances (cf. Norman, 1988).

The basic idea of the design strategies in the category *Force* is to compel the sustainable behaviour upon the users, through limited functionality or by restraining the undesired behaviour.

In the category *Match*, finally, the basic idea of the design strategies is that products and services should be adapted to behaviours that the user already performs, to the user's behavioural intentions, or to a behaviour already desired by the user.

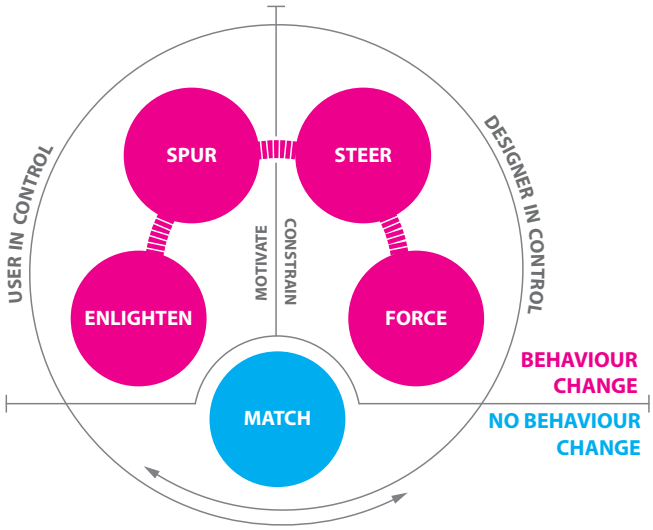


Figure 1. The model for categorising design strategies.

PHASE 2: IDENTIFYING BARRIERS

IDENTIFYING BARRIERS TO SUSTAINABLE BEHAVIOUR

Several studies suggest that every attempt to design for sustainable use should begin with a user study to identify barriers towards the target behaviour and to get to know the target group. This will enable designers to choose the design strategies that has the potential to deal with the identified barriers and that suit the target group the best.

One example is Tang and Bhamra (2009) who performed two pilot studies in order to improve energy efficiency of fridges and freezers. In the studies, they applied qualitative, user-centred, research techniques to investigate daily practices, needs, attitudes, intentions and actions of users. Their findings showed that an understanding of use behaviour provided directions for ways to design for sustainable behaviour. Another researcher, Manning (2009), stated that sustainable behaviour is most likely to arise when people encounter few barriers towards the sustainable action. Such barriers can be a lack of infrastructure, an extra expense, or a lack of knowledge, etc. Consequently, a successful sustainability campaign should start with an analysis on what barriers people face regarding the subject of the campaign (ibid.). Gardner and Stern (2002) strengthen this statement further in their recommendation that when creating an environmental programme, the numerous and varying barriers towards a sustainable behaviour should be identified, and that the best way to do this is by observing or interacting with the people whose behaviour is to be changed.

METHOD AND PROCEDURE

The method contextmapping – as described by Sleeswijk Visser, Stappers, van der Lugt and Sanders (2005) – was chosen for the exploration of the behaviour of dosing washing detergent.

Through observations, focus groups and questionnaires, an understanding of what people do, say and think can be achieved (Sanders and Dandavante, 1999). Yet, as found by Habib, El-Masri and Heath (2006) this may not reveal all the barriers towards correct dosing of cleaning agents. Instead, with the generative techniques employed in contextmapping, deeper insights in thoughts, feelings and dreams are claimed to be gained (Sanders and Dandavante, 1999).

Fifteen participants were recruited to the contextmapping study and three separate meetings were held with four to six participants in each. Altogether this study involved nine women between 26 and 60 years old (mean age=42), and six men between 25 and 63 years old (mean age=34).

A week prior to the meetings, all participants received a sensitising booklet consisting of a couple of self-reflective exercises on the topic of cleaning and personal hygiene. In the meeting the participants were presented with four statements regarding dosing and were asked to react to these statements by making collages and thereafter to explain their collages. The meeting ended with a group discussion on dosing of cleaning agents.

The discussions were recorded and transcribed. The comments representing barriers towards correct dosing of cleaning agents were identified and then sorted into themes where each theme represented one barrier to correct dosing.

RESULTS AND ANALYSIS

According to the results, the most prominent barriers to correct dosing of washing detergent were:

- **Habitual use of washing detergents -** The use of washing detergent was for many a habitual behaviour. Some of the participants were aware of this and that their use of washing detergents was based on assumptions and feelings rather than on facts and knowledge.
- **Need for feedback on dosing and cleanness -** There is no clear feedback on the amount of washing detergent used or on the cleanness of the laundry. The scent from the washing detergent sometimes served as an indirect feedback on how clean the laundry was.

- **Estimated dosing -** The participants had an unreflecting approach towards dosing. Many of the participants stated that they never measured the washing detergent. Instead they poured the detergent directly from the package and roughly estimated, guessed or thought they had an intuitive understanding of the amount needed. Most participants did not read the dosing recommendations on the packages, yet most of them thought that they dosed moderately. Additionally, many of the participants did not know the hardness of their water and did not know how this influences the amount of washing detergent needed.
- **Packaging as obstacle to moderate dosing -** The package of the washing detergent had an impact on the dosing as some packages are difficult to dose moderately from. The size of the package served as a dosing guide, especially for one participant who based the size of a reasonable dose of washing detergent directly on the size of the package. Several participants stated that they wanted dosing aids to be integrated in the package and one participant said he wanted an alternative where he does not have to think.
- **Desire for convenience -** Several participants expressed that they wanted the washing to go quickly. Other participants mentioned convenience and efficiency as important factors. These wishes may affect how a person doses washing detergent.

PHASE 3: CHOOSING STRATEGIES AND DEVELOPING PROTOTYPES

MATCHING STRATEGIES WITH BARRIERS

All identified design strategies for sustainable behaviour were reviewed and after discarding strategies that would be unsuitable for dosing of washing detergent a judgment was made regarding what specific barriers each of the remaining strategies would help overcome. The four strategies with the potential to bridge most barriers were chosen. An overview of what barriers the chosen strategies were thought to overcome is provided in Table 1. Each strategy resulted in one prototype, the ambition being that one prototype should influence the users according to only one of the strategies. Prototypes were designed (by the two first authors) and built to evaluate their functionality before the final prototypes were created.

Design Strategy --- Barrier	Eco-Affective Design	Competence and Autonomy	Scripting	Habit Intervention
Habitual use…	X		X	X
Need for feedback…		X		
Estimated dosing	X	X	X	X
Packaging as an obstacle			X	X
Desire for Convenience	X		X	X

Table 1. The barriers that the different strategies could help a user to overcome.

ECO-AFFECTIVE DESIGN

Eco-Affective Design is a strategy aiming for evoking negative emotional reactions to an undesired behaviour's environmental impact or positive emotional reactions regarding the desired behaviour's reduction of environmental impact. The strategy is categorised into the *Enlighten* category.

The prototype developed from the strategy was a measuring cup with a plastic frog on a stone. The level of the frog's feet represents a suitable dose for most washes (Figure 2). If more detergent is used, the frog is drowned in detergent. The idea is that the user will enjoy using the measuring cup because of the appealing frog but experiencing negative emotions when overdosing.

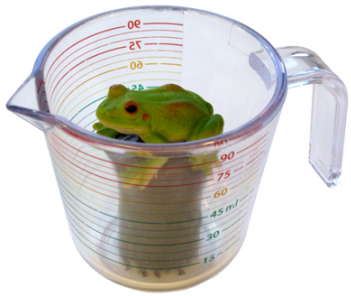


Figure 2. The prototype developed from the strategy Eco-Affective Design.

COMPETENCE AND AUTONOMY

The aim of the strategy *Competence and Autonomy* is to facilitate the formation of intrinsic motivation for an action through providing sufficient challenge for the user in order to support his or her feelings of being capable. According to the Self-Determination Theory (Deci and Ryan, 2000) motivation can be intrinsic or extrinsic. Intrinsically motivated behaviour takes place when the act in itself constitutes the reward and the behaviour is engaged in out of free will. Intrinsic motivation is closely linked to the needs for autonomy, competence and relatedness, and without these factors intrinsic motivation is less likely to arise (ibid.). The strategy *Autonomy and Competence* is categorised into the *Spur* category.

The prototype developed from the strategy was a dosing kit consisting of a wheel chart, a measuring cup and a laundry basket with a scale integrated into its handle (Figure 3). After weighing the laundry the user can set the wheel chart and dose accordingly. The idea is that the user should feel capable and competent because he or she now knows exactly how much washing detergent to use. As the user can do this on his/her own he/she will also feel autonomous.



Figure 3. The prototypes developed from the strategy Competence and Autonomy.

SCRIPTING

The idea behind the strategy *Scripting* is to design artefacts in a way that triggers sustainable usage, either by obstacles for unsustainable use or by making the sustainable behaviour so effortless that it becomes the natural way of usage (Wever, van Kuijk and Boks, 2008). *Scripting* is similar to the concepts of affordances and constraints as explained by Norman (1988), a similarity also noted by Lilley, Lofthouse and Bhamra (2005). The strategy is categorised into the category *Steer*.

For the prototype developed from the strategy, the functionality was copied from a package already available on the market (Figure 4). In the package the volume of detergent is constrained as only a predetermined amount will pour out. The package has to be tilted back and then forth again for a second dose. This way the unsustainable behaviour of 'thoughtless pouring' is constrained and the sustainable behaviour of dosing in a deliberate way is facilitated as the user can easily count the number of tilts.



Figure 4. The prototypes developed from the strategy Scripting.

HABIT INTERVENTION

One of the most effective ways to break a habit is to make the habit impossible (Jager, 2003). The idea behind the strategy *Habit Intervention* is to break an unsustainable habit in a way that the user approves of and, thereafter, to facilitate the formation of a new sustainable habit. Verplanken and Wood (2006) pointed out that it is important not only to disrupt an undesired habit but also to facilitate the formation of a new habit in order to ensure maintenance of the new behaviour. The strategy is categorised into the category *Force*.

The prototype developed from the strategy was washing detergent tablets in a tube (Figure 5). The intention was to break the previous habit by making it impossible to dose by estimations of the

size of the washing powder pile and, instead, to facilitate the formation of a new habit by introducing a simple cognitive script for the new behaviour, i.e. the user only had to choose between one and two tablets (depending on the dirtiness of the laundry and size of washing machine). The only information provided was dosing recommendations for the type of water in the region of the study. (Washing detergent tablets are not available in conventional stores where the study was made.)



Figure 5. The prototypes developed from the strategy Habit Intervention.

PHASE 4: EVALUATION OF PROTOTYPES

The strategies were evaluated in a field study.

METHOD AND PROCEDURE

Sixteen households were recruited, consisting of altogether 29 individuals; 16 women between 19 and 58 years old (mean age=36), and 13 men between 17 and 49 years old (mean age=31). None of the participants measured their washing detergent prior to the study, hence all of them potentially overdosed.

The study was run according to a between-subject design without control group, i.e. all participating households received one prototype and used the same prototype for the entire test period. The participants documented their individual dosing behaviour themselves by weighing the package of washing detergent before and after dosing and noting the weight in a booklet. This was done at three stages during a total period of five months; during a one month long period prior to receiving the prototypes (documented in booklet 1, baseline), during an obligatory one month long use phase in which the participants were obliged to use the prototypes (documented in booklet 2) and during an optional three months long phase in which the participants had access to the prototypes but were told to use the prototypes only if they wanted to (documented in booklet 3). If a refill of washing detergent was needed, new packages were delivered to the participants' homes throughout the total period.

After the last phase, in-depth interviews were conducted with each household. In the interviews the following topics were covered; the effectiveness of the prototypes in inducing the target behaviour and indications of long-term effectiveness, the acceptance of the prototypes and indications of long-term acceptance, the extent to which the applied strategy was accountable for the effectiveness and acceptability of the prototypes, and unintended influence of the study setup. The participants were also asked what they did in their everyday life for the sake of the environment and to characterise themselves according to an abbreviated version of Persson's and Hemberg's (2010) characters representing different attitudes to environmental issues. Any discreditable personality traits in the descriptions had been rephrased though to make all equally attractive.

All interviews were recorded and transcribed. The material was categorised according to the four topics covered in the interviews. The six different descriptions of attitudes to environmental issues were clustered according to pro-environmental engagement. This resulted in four degrees or 'levels'. The lowest level was assigned a score of zero points whereas the highest level was assigned three points. A similar system was constructed for the participants' descriptions of their pro-environmental actions. Then, the score for pro-environmental engagement and the score for stated pro-environmental actions were summed up to a total score that was used to 'measure' environmental concern. This total score thus ranged from zero to six points (in Figure 6 called eco-points). In addition, the data from all booklets were compiled and the weight of the doses reported in grams was converted into 'volume'.

RESULT AND ANALYSIS

ECO-AFFECTIVE DESIGN

When in use, the effectiveness of the frog cup seemed high, as all participants either lowered their consumption of washing powder or remained at an already modest level (Figure 6). Two participants stopped using the frog cup after the obligatory use phase but continued to dose moderately. Of the nine participants who were present at the closing interviews, five participants mentioned that they would like to keep the prototype and all participants except one expressed that they wanted to continue with a low consumption

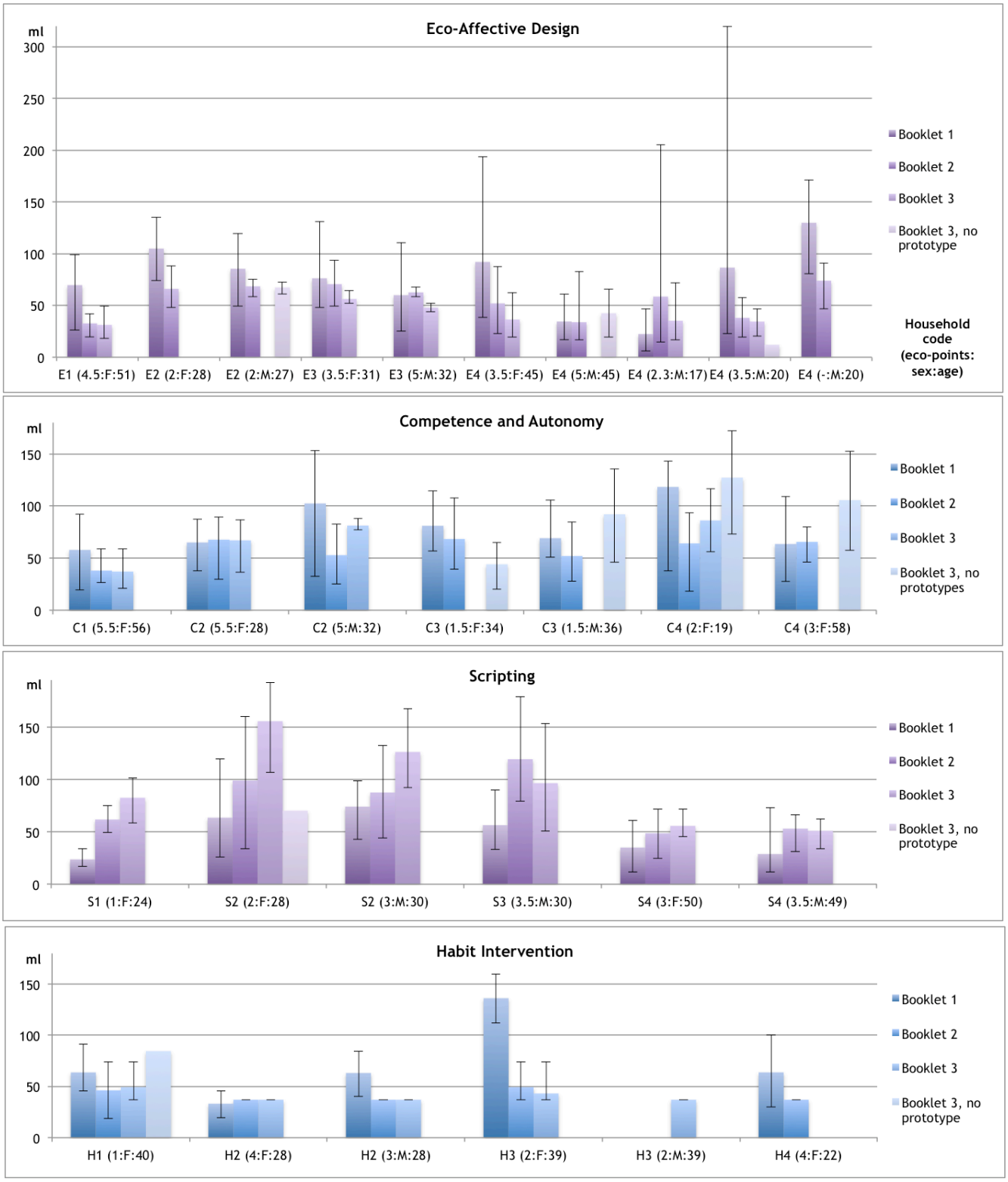


Figure 6. The dosing behaviour of the participants represented by mean values from the three booklets. The error bars show the highest and the lowest dose in each booklet. The participants are named according to the following: household code (eco-points:sex:age).

of washing detergent. These statements indicated a potential for long-term effectiveness. The interpretation is that the participants were enlightened about the importance of moderate dosing and motivated to act upon this attitude. The acceptability of the frog cup varied between households but was in general 'fair', as most participants expressed a certain liking for the frog. The largest obstacle for long-term acceptance was the opinion that the frog was "… in the way …", i.e. an obstacle when pouring detergent into the cup. This aside, the participants' acceptance seemed to remain unchanged or even grow stronger during the study.

All but two of the participants mentioned that they had associated the frog cup with environmental aspects, thus it seems as though the prototype acted as a reminder of environmental issues, helping the participants to link overconsumption of washing powder with environmental pollution. The metaphor of the frog being suffocated or drowned by extensive dosing was however only explicitly mentioned by two participants. Other participants mentioned feelings of guilt or positive feelings of doing good for the environment, but without relating it to the frog. Nevertheless, the participants using the frog cup mentioned the environment more frequently than did the participants who tested other prototypes. This could explain why the participants were motivated to dose moderately but not how they determined what a correct dose was but when using the frog cup it could feel awkward to cover the frog with detergent. Thus, a possible answer is that *Scripting* with semantic constraints (cf. Norman, 1988) may have played a role in the sense that one does not want to make the main character, i.e. the frog, disappear. In addition, the scale from green to red (Figure 2) can be seen as *Scripting* with cultural constraints (ibid.), as green denotes good/okay/correct in many cultures while red denotes ad/error/warning.

Competence and Autonomy

The effectiveness of *Competence and Autonomy* was very high when the prototypes were in use. Nevertheless, four out of seven participants stopped using the kit after the obligatory use phase and three of these participants increased their mean doses compared with their already high mean doses during baseline (Figure 6). Therefore, the potential for long-term effectiveness must be regarded as low. One of the participants that increased her mean dose after ceasing to use the kit stopped using it because she thought (falsely) that she had learnt how to dose moderately without any dosing device. This indicates that the dosing kit made her feel competent – but it did not make her competent enough. However, for some participants the long-term efficiency seemed higher. One example is a participant who after the withdrawal of the prototypes started to use another measuring cup but remembered the recommendations on the wheel chart and continued to dose accordingly. In addition, the acceptability of the laundry basket was very low. The most prominent problem was that it could not stand upright without support and for some participants this was the reason for their ceased use of the kit. On the other hand, the acceptability of the wheel chart and the measuring cup was generally high.

Feelings of competence or autonomy were not triggered to the intended extent. The wheel chart was, at least in the beginning, enjoyable to some participants and appeared to induce some feelings of competence. The accuracy in measuring detergent as reported by all of the participants indicated that they acknowledged the importance of correct dosing and many of the participants enjoyed getting to know the weight of the laundry. Nevertheless, to most of the participants the occasional feeling of competence when using parts of the kit was not strong enough to create a motivation to use the whole kit, possibly as any positive feelings were overridden by aversion towards the laundry basket.

Two out of all the participants in the study scored 5.5 eco-points (of 6) and, by pure chance, both these participants got the *Competence and Autonomy* prototype (Figure 6). These two individuals were also the most persistent users of this kit, which could be explained by their attitude.

Scripting

All the participants in the *Scripting* group dosed according to recommendations – or even less also before introduction of the prototypes. Using the prototypes, all participants continued to dose according to recommendations or below, but some of the packages had technical flaws and the amount of washing detergent per tilt was too high, resulting in much higher doses when using the prototypes than those measured during baseline conditions (Figure 6). Yet, as the participants dosed according to the recommendations the effectiveness in inducing the target behaviour was considered high and would have resulted in maintained or lowered doses if the packages had functioned as intended.

The potential for long-term effectiveness appeared to be good as all participants continued to use the prototypes and to dose according to recommendations throughout the study. However, it is uncertain if the moderate dosing would continue without sustained access to the prototype packages. One of the participants had started to use a measuring cup when his last prototype package was finished, and two participants expressed a desire to continue to dose a little less even if they planned to start pouring directly from the packages again. The remaining three participants did either not reveal any thoughts on how they would dose in the future, or would return to their previous behaviour. None of the participants said that they would start buying the *Scripting* package when they learnt that packages with the same functionality were available in stores. In total, the acceptability of the prototypes seemed fair but not exuberant, yet the packages were accepted enough by the participants to make all of them use the packages also throughout the optional use phase. All participants appeared to appreciate the functionality and the convenience of the integrated dosing system, even if two participants found it a bit tiresome to tilt the package back and forth.

The design strategy *Scripting* played a role in the effectiveness of the prototypes in three of the households as the constraint made the participants dose their detergent in a more conscious manner instead of just pouring an estimated amount of detergent from the package. Thus the tilting back and forth appeared to be a suitable physical constraint (cf. Norman, 1988). For the fourth household their initial moderate dosing was the reason for them to dose according to the instructions. Regarding acceptability the households seemed to appreciate that no mental effort was needed for handling of the product, indicating that they liked to be directed.

Habit Intervention

When being used, the effectiveness of the tablets in inducing the target behaviour both during the test period and long-term seemed very high (Figure 6). Still, the participants reported that they most probably would return to their initial dosing routines even if one participant mentioned that he would start looking for washing detergent tablets in the stores. Additionally no requests for dosing devices were noticed. This indicates that the potential for long-term effectiveness without sustained access to the prototypes was low. Regarding the level of acceptability the results were ambiguous. Three participants immediately accepted the product and for them the dosing quickly became habitual. A fourth participant reached a high acceptance after a while. The acceptability was related to an appreciation of being steered as the participants who showed a high acceptance liked not to have to make decisions about dosing. The tablets were considered handy and easy to use and it is noteworthy that the participant with the greatest reduction in mean dose was very satisfied with the tablets. However, one of the participants disliked the tablets partly because she could not adapt her dosing as she wanted, i.e. because she was steered. Another participant was very sceptical towards trying new things in general and had strong preconceptions regarding washing detergent. She did not want to try the tablets and her acceptance was very low. None of the participants that tested the tablets were bothered by the fact that they were adapted only for soft water (the type of water in the region of the study). This indicates that the participants accepted this type of steering.

The strategy *Habit Intervention* consists of two phases; intervention of the previous habit and formation of the new habit. Due to the study design, the effectiveness of the tablets in the first phase is hard to judge as the participants initially were obliged to use the tablets. The tablets introduced a new way of measuring washing detergent, by number of tablets and not by estimating the size of the detergent pile. In this way the prior habit was made impossible. For one of the households this was important as their initial doses were very high. They doubted that one tablet would be enough for one wash but still used only one tablet. If they had received the *Scripting* prototype, it is possible that they would have added another dose (i.e. a tilt) just to "make sure" as they would then have been able to compare with their prior doses. The finding that the participants most probably would return to their initial dosing routines is in line with the nature of the strategies in the category *Force* as they serve the purpose to steer and not to motivate.

DISCUSSION

The study presented in this paper could be described as an explorative study involving a limited number of participants and, as with any study, the outcome is partly a result of the study design. It is possible that the participants gained an increased awareness of their washing detergent dosing behaviour and of the importance of moderate dosing by merely participating in the study and by filling in the booklets. It is also possible that this increased awareness had an affect on their dosing behaviours during baseline conditions as well as during the remaining part of the study. Nevertheless, it was not possible to understand how to dose in order to dose moderately from just filling in the booklets. The participants would still need to, for instance, figure out the water hardness in the region. Thus, filling in the booklets and participating in the study as such may have increased the participants' motivation to dose moderately but it did not increase their ability to dose moderately. As suggested in the Motivation-Ability-Opportunity-Behaviour Model, ability and opportunity are important for creating a link between motivation and behaviour (Ölander and Thøgersen, 1995). Nevertheless, the potentially increased motivation may have resulted in a more positive response to the prototypes.

Another factor to influence the results is the choice of participants. The participants were all 'estimate' users of washing detergent, i.e. they did not measure when dosing. Yet, as revealed in the baseline booklet, not all participants overdosed. For several of the participants no reduction of mean dose per wash was necessary and their changes in dosing behaviour were also less than for those other participants that overdosed significantly. Another factor that could have influenced the outcome of the study is the participants' knowledge of laundering. However, in the closing interviews, no prominent difference could be noted between the participants. A third factor to be considered is attitude towards environmental issues. Provided the small number of participants in the study it is not possible to draw any general conclusions but it is reasonable to assume that a strong pro-environ-mental attitude would increase the motivation to adopt a solution that would help maintain a sustainable behaviour. For instance, the participants differed in attitude and the two participants scoring high regarding environmental concern were also the most persistent users of the *Competence and Autonomy* kit.

Regarding the outcome of the study, some of the participants used the prototypes throughout the study while others ceased their usage directly after the obligatory use phase. As a result, actual behaviour after ceased use of the prototypes could be studied only in a few households. In the remaining households the participants' thoughts on how they would dose after withdrawal of the prototypes was the only indicators of what would happen. Some of the participants said that they would try to continue to dose moderately and considered to acquire a measuring cup while others stated that they would return to estimating doses as they did before the study. This indicates that the long-term effectiveness of the prototypes cannot be ensured without sustained access, a finding that corresponds with earlier findings from

the area of design for sustainable behaviour. For instance Dwyer et al. (1993) reviewed several studies on feedback and in none of the studies the sustainable behaviour achieved during the time feedback was given had been maintained after the feedback intervention ended – this even though some feedback interventions lasted over two years! Therefore, it is here suggested that behaviour steering elements should be integrated parts of products and services and not separate features used occasionally.

Amongst the participants that used the frog cup all except one said that they wanted to continue with a low consumption of washing detergent. The motivation to dose moderately also after the study was higher in this group than in the other groups. The number of participants was too small to determine if this was caused by the characteristics of the prototypes or only by individual differences, but according to the model for categorisation of design strategies only the strategy of *Eco-Affective Design* aims at motivating the users to dose moderately by enlightening them, whereas the aims of the influencing techniques of *Scripting* and *Habit Intervention*, placed in the categories *Steer* and *Force*, is to direct, not to motivate the users.

Given the results, all strategies could be considered efficient when the prototypes were used. The major difference was instead the difference in acceptance. Apart from flaws of the prototypes (due to deficient manufacture), two important factors for low acceptance were found. The first one was that the use of the prototypes required some kind of effort, exemplified by the slight disapproval of the tilting of the *Scripting* package and the low acceptance of the *Competence and Autonomy* dosing kit, especially of the bothersome laundry basket. It has earlier been proposed that high effort can lead to amotivation to perform sustainable behaviours (Pelletier, Dion, Tuson, and Green-Demers, 1999). If applied to the *Competence and Autonomy* dosing kit this indicates that the effortful handling might have created amotivation, thus undermining the attempts to create motivation by designing for feelings of competence.

The second factor for low acceptance was disapproval of being steered by the prototype, exemplified by the *Habit Intervention* prototype where one participant expressed disliking, at least partly because of the steering element. According to Lilley (2009) people are used to master their products and are not used to the products steering them. On the other hand, a majority of the participants that used the prototypes of *Scripting* and *Habit Intervention* appreciated the very same feature, i.e. they wanted to be steered. It is not established how this finding corresponds to other studies and there is thus a need for further investigations on in what areas direction of behaviour is acceptable and in what motivation is advisable (cf. Lilley, 2009). Yet, it is reasonable to assume that people, depending upon application area, show different level of receptiveness towards different design strategies. It is hence possible that a combination of design strategies should be used. With a combination of design strategies it is also possible to address different barriers with the same artefact.

Finally, it was found that high effectiveness can be achieved together with high acceptance, also when changing from a very high consumption to moderate dosing. This implies that designers can motivate and direct sustainable behaviour beyond the possibilities of laws and regulations. A key issue for acceptance might be that the product support, and not contradict, the user's values, as suggested by the study of Lilley (2009).

CONCLUSION

The aim of this study was to examine the long-term acceptability and effectiveness in inducing sustainable behaviours of four different design strategies. Based on the results of the study the following conclusions can be drawn.

- The effectiveness of the frog cup was high, yet the effectiveness of the strategy *Eco-Affective Design* was not fully understood as other strategies also played an important role in the inducement of the behaviour, for instance scripting with semantic and cultural constraints.

- The potential for the strategy of *Competence and Autonomy* was difficult to establish. It appears challenging to design for strong feelings of competence and autonomy and at the same time create a product or service that requires little effort to use.
- The strategy of *Scripting* appears promising as effectiveness was high and acceptability seems to have the potential to be high.
- The strategy of *Habit Intervention* showed ambiguous results, as prior habits were not always disrupted by the new habit. However, when the prior habit was disrupted, *Habit Intervention* seemed promising and the formation of a new habit through *Scripting* was successful.

Thus, product design can be a feasible way to induce sustainable behaviour and high acceptance can be achieved together with high effectiveness. It is advisable, though, that the inducing element is an integrated part of the product or service for sustained behaviour changes.

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